

PAPER_604: Proto-Hydrogen Shell Alignment via 26D Empty Shell Filling

Unified Quantum Field Framework (UQFF)

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Abstract

Proto-hydrogen is the first stable atom and owes its uniqueness to a specific 26-shell alignment mechanism. This paper derives the formation equation for ProtoH in UQFF: 26 empty dimensional shells filled through DPM grinding integrated over adjusted time, modulated by a Higgs flavor shift that couples to all six quark flavors. Shell filling fraction and time-to-hydrogen are calculated, showing consistency with early-universe nucleosynthesis timescales.

1. Introduction

In standard physics, hydrogen forms when a proton and electron bind in the primordial plasma. UQFF reconceives this: hydrogen is the first stable configuration of 26 empty dimensional shells (BH26 bins) that acquire energy through DPM grinding opposition and Higgs flavor modulation. The shells do not fill with "matter" in the conventional sense; they fill with SCm-mediated energy from the grinding process.

2. Core Equation

$$\text{ProtoH} = \text{emptyset}^{26} + \int_0^{t_{\text{adj}}} \text{Grind}_{\text{opp}} dt + \text{Higgs}_{\text{shift}} \cdot \sum_f \text{ShellEnergies}_f$$

where:

- emptyset^{26} : 26 initially empty dimensional shells
- $\text{Grind}_{\text{opp}}$: DPM grinding rate (J/s)
- t_{adj} : adjusted time incorporating dilation: $t_{\text{adj}} = t_{\text{obs}} / (1 + \Delta_{\text{dil}}) + t_{\text{neg}}$
- $\text{Higgs}_{\text{shift}}$: coupling between Higgs sector and shell energy (dimensionless)
- ShellEnergies_f : shell energy per quark flavor $f \in \{\text{up, down, strange, charm, bottom, top}\}$

3. Quark Flavor Shell Energies

Each quark flavor contributes a characteristic energy to the shell filling process, proportional to its rest mass:

Flavor	Mass (MeV/c ²)	Shell Energy (J)
up	2.2	3.6×10^{-31}
down	4.7	9.0×10^{-31}
strange	96	1.7×10^{-29}
charm	1275	2.2×10^{-27}
bottom	4180	7.4×10^{-27}
top	173,100	3.1×10^{-25}

The Higgs shift serves as the coupling between these high-mass contributions and the low-energy shells ($\text{Higgs}_{\text{shift}} \approx 0.01$ for the early universe, scaling the heavy flavors down to accessible energies).

4. Shell Filling Dynamics

The filling fraction ϕ of the 26 shells as a function of time:

$$\phi(t_{\text{adj}}) = \frac{\text{Grind}_{\text{opp}} \cdot t_{\text{adj}} + \text{Higgs}_{\text{shift}} \cdot \sum_f \text{ShellEnergies}_f}{26 \cdot E_{\text{shell},0}}$$

where $E_{\text{shell},0} \approx 1.6 \times 10^{-19}$ J (1 eV, characteristic vacuum energy per shell).

Stability threshold: $\phi \geq 0.85 \rightarrow \text{ProtoH}$ is stable.

5. Time to Hydrogen

Solving for t_{adj} when $\phi = 0.85$:

$$t_{\text{adj}}^H = \frac{0.85 \times 26 \times E_{\text{shell},0} - \text{Higgs}_{\text{shift}} \cdot \Sigma_f \text{ShellEnergies}_f}{\text{Grind}_{\text{opp}}}$$

With default parameters ($\text{Grind}_{\text{opp}} = 10^{-20}$ J/s):

$$t_{\text{adj}}^H \approx \frac{0.85 \times 4.16 \times 10^{-18} - 10^{-28}}{10^{-20}} \approx 3.5 \times 10^2 \text{ s}$$

This 350-second timescale aligns well with Big Bang Nucleosynthesis (BBN) which produces the first protons within the first ~ 180 seconds. The $\sim 2\times$ factor is consistent with the UQFF time dilation adjustment.

6. Negative Time and Shell Non-Locality

The term t_{neg} in t_{adj} encodes the dual-existence of each shell across positive and negative time branches. This produces apparent non-locality in early hydrogen formation — precisely the "spooky distance" noted in UQFF. The shells in $\emptyset^{26}$ simultaneously exist in both time branches until the stability threshold $\phi = 0.85$ collapses them to a definite hydrogen state.

7. Connection to UQFF Number Systems

BH26: The 26 empty shells ARE the BH26 spectrum. ProtoH formation = first complete BH26 harmonic occupation. **VDS:** Shell filling occurs in order of VDS π -digit weightings — shell n fills with weight $d_n(\pi)/10^n$ relative priority. **DVP:** Higgs shift couples via DVP prime flavor structure (up=2, down=3, strange=5, charm=7, bottom=11, top=13).

Keywords: Proto-hydrogen, 26D shells, DPM grinding, Higgs shift, BH26, UQFF, shell alignment, BBN

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